

Carbon Emissions and Basic Carbon Management for SMEs

🕒 15 min · Supporting learning resource (LSM)

Module 4 · Resource Efficiency: Energy, Carbon and Water
Unit 3 · Carbon Emissions and Basic Carbon Management



Introduction

Carbon emissions are increasingly discussed in relation to climate change, sustainability and business responsibility. For many SMEs, however, the concept of carbon emissions may still appear abstract or disconnected from daily operations.

This Learning Support Material complements the video lesson by summarising the key concepts related to carbon emissions and their relevance for SMEs. It helps learners understand where emissions typically come from in business activities and supports reflection on simple, non-technical actions that can contribute to emission reduction and improved efficiency.

Key concepts explained

What carbon emissions are

Carbon emissions refer mainly to carbon dioxide released into the atmosphere when energy is used. This occurs, for example, through electricity consumption, heating, transport or the operation of machinery and equipment.

For SMEs, carbon emissions are closely linked to everyday activities. Whenever energy is consumed, emissions are likely to be generated, even if this is not immediately visible.

Carbon emissions and climate change

Carbon emissions contribute to climate change by increasing the concentration of greenhouse gases in the atmosphere. Climate change can affect businesses through rising energy costs, changing regulations and increasing expectations from customers and partners. Understanding this link helps SMEs recognise that carbon emissions are not only an environmental issue, but also a factor influencing long-term business performance and resilience.

Main emission sources in SMEs

In most SMEs, the main sources of carbon emissions are related to energy use and transport. Typical sources include electricity consumption, heating and cooling systems, company vehicles and production or service-related processes.

Identifying which activities generate emissions is an important first step towards reducing them.

Why this matters for SMEs



For SMEs, carbon emissions are closely connected to costs, efficiency and daily operations:

- higher energy use usually means higher emissions and higher costs
- inefficient processes increase both environmental impact and operating expenses
- reducing emissions often goes hand in hand with improved efficiency

By becoming aware of their emission sources, SMEs can identify opportunities to reduce costs while supporting more sustainable business practices.

Practical examples or typical situations

SMEs may recognise emission-related issues in situations such as:

- frequent use of company vehicles for short trips
- heating or cooling systems operating inefficiently
- energy-intensive equipment used without clear necessity

These examples illustrate how carbon emissions are often linked to routine practices rather than exceptional activities.

Key takeaways

- Carbon emissions are mainly linked to energy use in SMEs
- Everyday activities contribute to emissions, often without being noticed
- Reducing emissions often improves efficiency and lowers costs
- Awareness is the first step towards basic carbon management

Reflection prompts

Take a moment to reflect on your own organisation:

- Which activities are likely to generate the most carbon emissions?
- How is energy used in daily operations?
- Which practices could be reviewed to reduce emissions and improve efficiency?

Link to next unit

This unit focused on understanding carbon emissions and their relevance for SME operations. In the next subunit, the module will continue exploring practical approaches to resource efficiency, building on energy and carbon awareness towards more comprehensive sustainability actions.

References

- Toolkit chapters: Energy and carbon introductory sections
- Pedagogical design reference: PA4 – Module 4, Subunit 4.3